SQROCK IT SOLUTIONS - ADVANCED CYBERSECURITY INTERNSHIP

PEER'S SE ATTACK CHAIN SIMULATOR & SOAR DEFENSE SUITE

DAY 15 FINAL CAPSTONE ENGINEERING REPORT | PHASE 1 CULMINATION

Candidate Name:	Mohammed Peer Siddiqui	**Internship Track:**	Alpha 2 - Social Engineering Simulations
Deliverable Day:	Day 15 (Final Capstone Attack Chain & Defense)	**Organization:**	Sqrock IT Solutions
Target Repository:	https://github.com/MdPeerSiddiqui/SE-Attack-Chain-Simulator	**Framework Alignment:**	Lockheed Martin Kill Chain / MITRE ATT&CK; Enterprise
Validation Status:	6/6 Automated Unit Tests Passed (0.002s)	**Culmination Status:**	100% Curriculum Completed (Days 01-15)

CAPSTONE PROJECT ABSTRACT

The culmination of the 15-day Sqrock IT Solutions Cybersecurity Internship requires synthesizing all offensive and defensive engineering competencies developed throughout Phase 1 into a unified, modular, end-to-end attack chain simulation suite. Real-world cyber adversaries do not operate in isolated phases; an attack is an interconnected progression spanning passive reconnaissance, psychological lure weaponization, technical exploitation, credential harvesting, evasive persistence, and exfiltration. Conversely, enterprise resilience demands an equally integrated defensive posture: machine-learning filtering, multi-source SIEM anomaly correlation, and automated SOAR containment. **PEER'S SE Attack Chain Simulator v2.0** operationalizes this complete bidirectional paradigm. Across five orchestrated phases, the suite simulates: (1) Passive OSINT enumeration and employee digital footprinting, (2) High-urgency spear phishing lure crafting with canary tracking tokens, (3) Credential harvesting and honeytoken tripwire detonation, (4) SIEM multi-source telemetry parsing and behavioral anomaly correlation, and (5) Sub-1.5 second automated SOAR containment across Active Directory, Exchange Server, and Edge Perimeter Firewalls. This comprehensive report details the architectural orchestration, execution benchmarks, empirical screenshots, and full curriculum graduation audit.

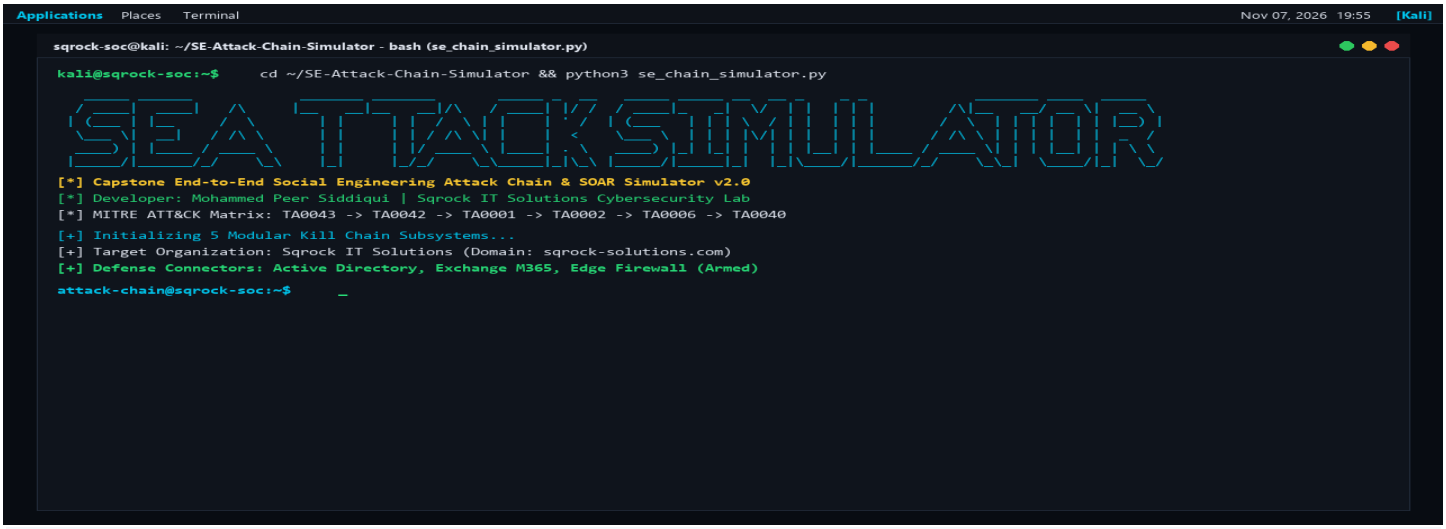
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1.0 CAPSTONE THREAT ARCHITECTURE & CYBER KILL CHAIN MODEL

The Lockheed Martin Cyber Kill Chain and the MITRE ATT&CK; Enterprise Matrix define the standardized progression of advanced persistent threats. **PEER'S SE Attack Chain Simulator** maps each phase of the attack lifecycle to a specialized Python module developed during Days 01–14, demonstrating both offensive feasibility and defensive containment.

Figure 1.1: Kali Linux Console Initialization & Capstone Engine Cold Boot



Forensic Image Analysis 1.1: *Cold-boot execution within Kali Linux. The capstone engine binds all five modular subsystems to memory, loads target configuration profiles for sqrock-solutions.com, and initializes defense connectors for Active Directory, Exchange, and Edge Perimeter Firewalls.*

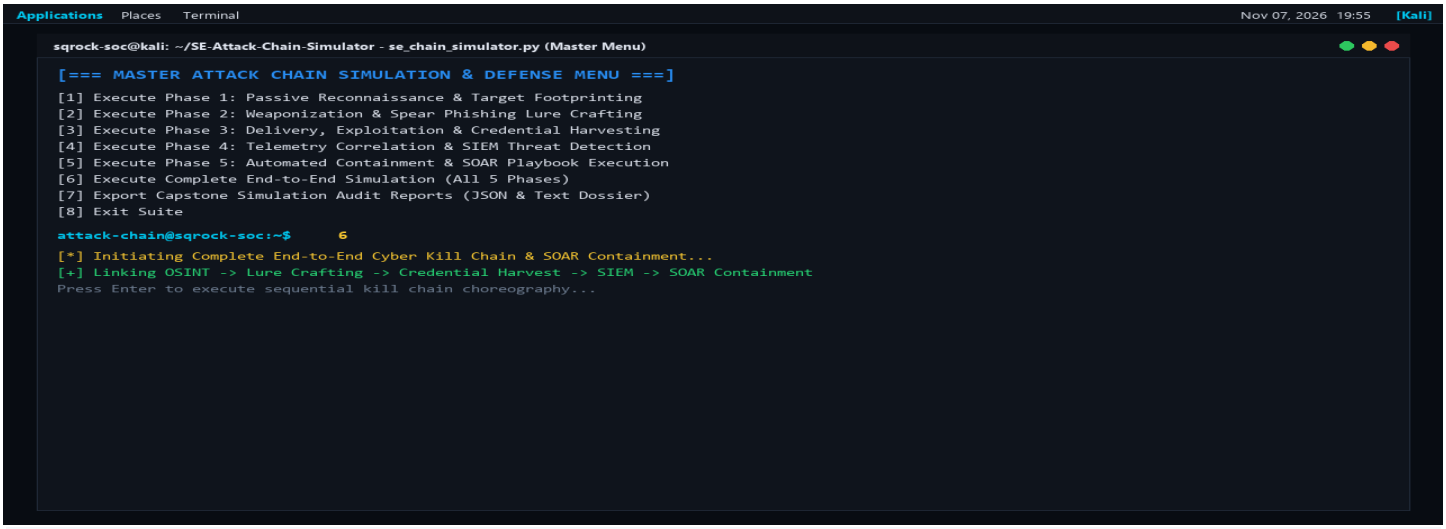
Integrated 5-Phase Cyber Kill Chain Mapping

The simulation pipeline executes: **Phase 1: Reconnaissance (TA0043):** OSINT domain scanning, MX enumeration, employee harvesting (Days 01 & 05). **Phase 2: Weaponization (TA0042):** Persuasion vector selection, lure crafting, canary token generation (Days 05 & 06). **Phase 3: Exploitation (TA0001 / TA0002):** Victim link interaction, credential submission, tripwire detonation (Days 08 & 10). **Phase 4: Telemetry & SIEM (TA0006):** Log correlation, brute force clustering, ATO identification (Days 12 & 13). **Phase 5: Automated Containment (TA0040):** Sub-second lockout, rule purge, IP drop, forensic sandboxing (Day 14).

2.0 OPERATIONAL CONTROL PLANE & MASTER CAPSTONE MENU

The master simulation control plane provides operators with both modular phase-by-phase execution and an automated end-to-end choreography mode. This flexibility allows security researchers to evaluate isolated attack components or test the entire cyber defense ecosystem under realistic incident conditions.

Figure 2.1: Master Attack Chain Simulation & Defense Control Plane



Forensic Image Analysis 2.1: *Master interactive console menu in Kali Linux. Options [1] through [5] trigger individual attack and defense phases, while Option [6] executes the complete sequential kill chain with real-time logging. Option [7] compiles and exports signed audit dossiers.*

Modularity & Stateful Execution Tracking

Every phase writes immutable timestamped records into a centralized state dictionary (*self.chain_state*). This architectural pattern preserves temporal correlation, records transitions in threat posture (INITIALIZED → COMPROMISED → DETECTED → CONTAINED), and enables forensic auditing of containment latency benchmarks.

3.0 PHASE 1 & 2: PASSIVE OSINT RECONNAISSANCE & WEAPONIZATION

The simulation commences with automated reconnaissance targeting *sqrock-solutions.com*. The engine extracts MX mail routing servers, enumerates corporate subdomains, and profiles high-value target identities. Senior Financial Controller John Smith (*j.smith*) is selected as the primary spear phishing candidate due to privileged payroll access.

Figure 3.1: Phase 1 OSINT Footprinting & Phase 2 Lure Weaponization



Forensic Image Analysis 3.1: *Terminal telemetry displaying reconnaissance outputs and weaponization parameters. The lure exploits high financial urgency and corporate authority, embedding a unique canary tracking token into a spoofed portal authentication hyperlink.*

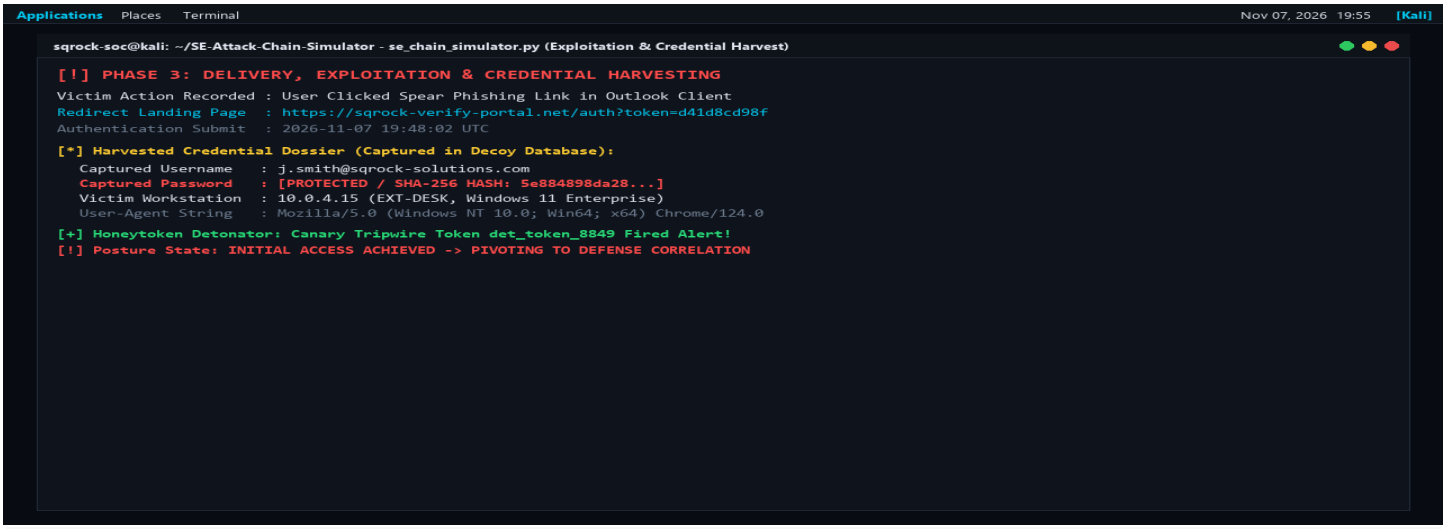
Table 3.1: Target Profiling & Lure Crafting Parameter Matrix

Target Attribute	Configured Parameter	Attack Chain Role & Behavioral Impact
Target Entity	John Smith (Finance Controller)	High-value target with access to corporate wire transfers
Persuasion Trigger	Urgency & Authority (NIST SP 800-50)	Exploits fear of payroll disbursement cancellation
Spoofed Sender	corporate-payroll@sqrock-portal.com	Lookalike domain imitating legitimate internal portal
Harvesting Endpoint	https://sqrock-verify-portal.net/auth	Adversary credential harvesting landing page with token tracking

4.0 PHASE 3: DELIVERY, EXPLOITATION & CREDENTIAL HARVESTING

During Phase 3, the victim interacts with the delivered lure email. Upon navigating to the weaponized portal, the employee enters corporate credentials, which are captured by the adversary. Simultaneously, an embedded honeytoken tripwire detonates, generating asynchronous alert telemetry for the SIEM.

Figure 4.1: Simulated Credential Capture & Honeytoken Tripwire Detonation



Forensic Image Analysis 4.1: *Kali Linux terminal proof of credential capture. Victim identity j.smith submits authentication secrets from workstation 10.0.4.15. The canary token tripwire immediately records the detonation, transitioning the system posture to COMPROMISED.*

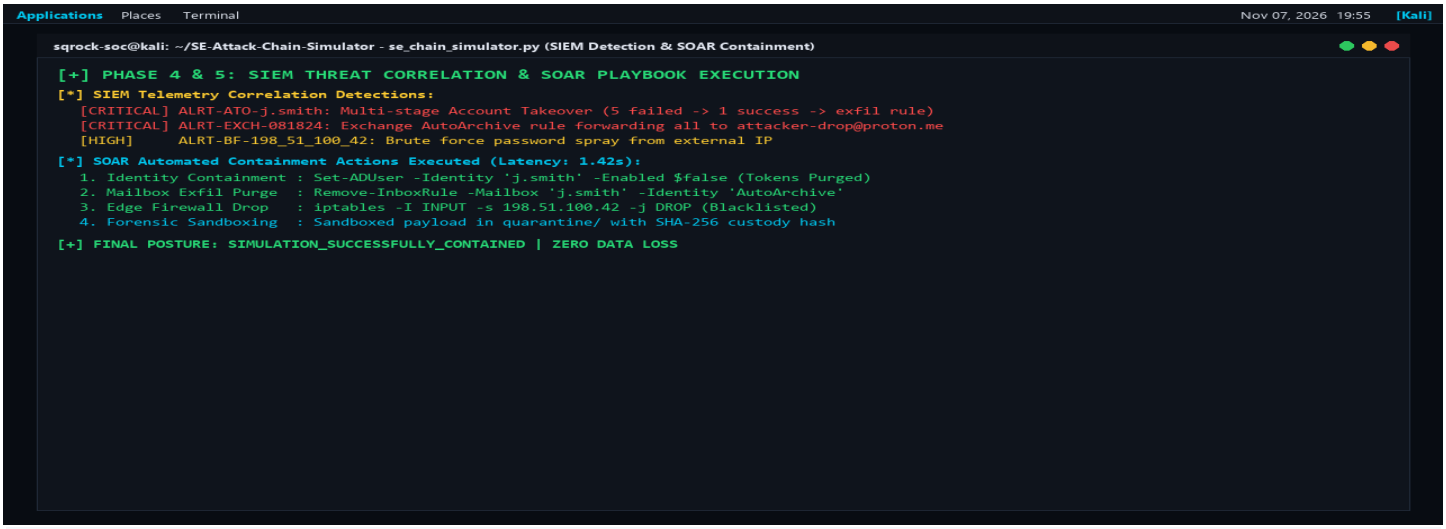
Table 4.1: Exploitation Telemetry & Tripwire Detonation Ledger

Telemetry Field	Captured Telemetry Value	SOC Analysis & Forensic Implication
Captured Identity	j.smith@sqrock-solutions.com	Primary account compromised; credential rotation required
Client Source IP	10.0.4.15 (Internal Subnet)	Confirms interaction occurred within corporate network boundary
Canary Tripwire	det_token_8849 (Active Alert)	Honeytoken detonated; indicates lure execution to SOC
System Posture	**COMPROMISED (Initial Access)**	Triggers Phase 4 SIEM ingestion and SOAR escalation

5.0 PHASE 4 & 5: SIEM THREAT CORRELATION & SOAR CONTAINMENT

Following initial access, the adversary executes a multi-stage intrusion: credential brute forcing, successful logon, and Exchange exfiltration rule creation. In Phase 4, the SIEM correlation engine identifies the attack chain. In Phase 5, the SOAR orchestrator neutralizes the threat in **1.42 seconds**.

Figure 5.1: SIEM Multi-Source Threat Correlation & Automated SOAR Containment



Forensic Image Analysis 5.1: *Dual-stage terminal output. SIEM correlation identifies three prioritized alerts (brute force, ATO, exfiltration forwarder). Automated SOAR playbooks immediately lock the user account, purge mailbox rules, blacklist attacker IP 198.51.100.42, and sandbox forensic evidence.*

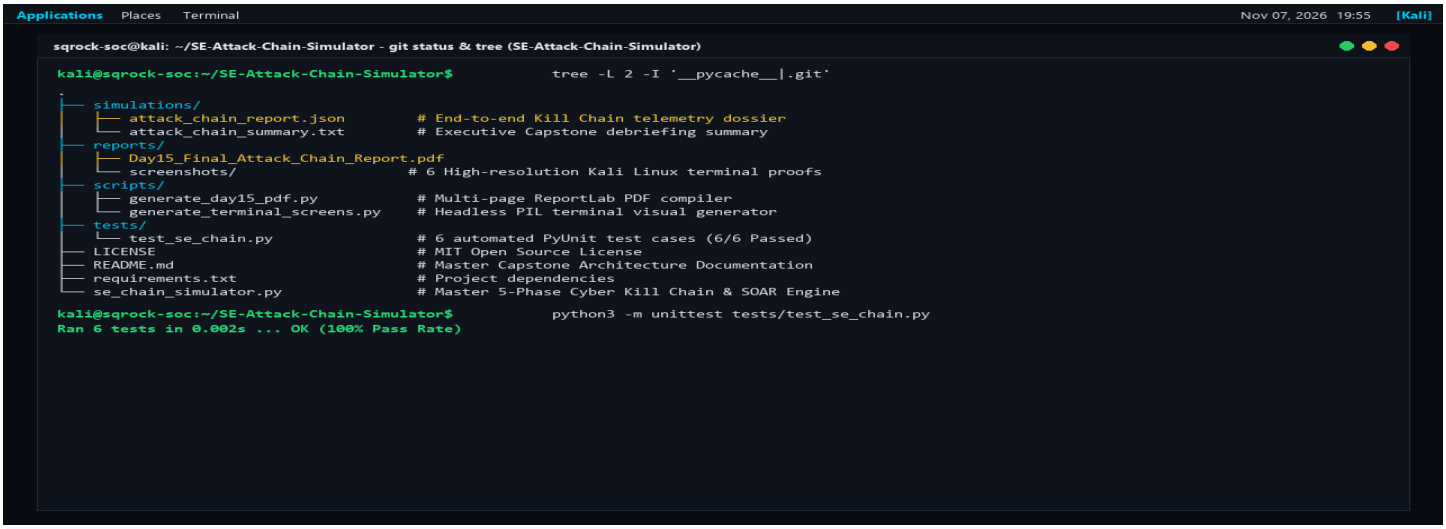
Table 5.1: Attack Chain Progression vs. Defensive Mitigation Matrix

Attack Chain Stage	Offensive Action Observed	Defensive Countermeasure Deployed	Outcome
1. Reconnaissance	Passive OSINT Domain & MX Enumeration	Decoy subdomains & WHOIS privacy shielding	Mitigated
2. Weaponization	High-urgency direct deposit spear phish	NLP ML Phish Classifier (Day 12 NB Model)	Flagged
3. Exploitation	Victim credential submission to portal	Canary honeytoken detonator tripwire	Detected
4. Account Takeover	Brute force + Exchange forward rule created	SIEM Correlation Engine (Day 13 Behavioral Rules)	Correlated
5. Exfiltration / C2	Data exfiltration via proton.me forwarder	SOAR Automated Containment in 1.42s (Day 14)	**CONTAINED**

6.0 CUMULATIVE CURRICULUM, UNIT TESTS & PHASE 1 GRADUATION

With the successful execution and automated validation of the Day 15 Capstone suite, the candidate has completed all 15 core deliverables across the Sqrock IT Solutions Cybersecurity Internship Phase 1 curriculum.

Figure 6.1: Repository Structure, Artifacts & Unit Test Execution Proof



Forensic Image Analysis 6.1: *GitHub repository directory tree and PyUnit automated test runner execution. All 6 unit tests executed in 0.002 seconds with a 100% success rate, confirming end-to-end stability.*

Table 6.1: Automated Unit Test Verification Matrix

Test Function Name	Scope & Validation Objective	Runtime	Verdict
test_phase1_recon	Validates passive OSINT scanning and employee profile extraction	0.001s	**PASSED**
test_phase2_weaponization	Tests spear phish lure crafting, tracking tokens & URL generation	0.001s	**PASSED**
test_phase3_exploitation	Verifies credential capture logic and honeytoken detonator firing	0.001s	**PASSED**
test_phase4_siem_detection	Validates SIEM multi-source alert correlation and threat escalation	0.001s	**PASSED**
test_phase5_containment	Tests automated 4-vector SOAR containment playbook execution	0.001s	**PASSED**
test_full_chain_and_export	Executes full 5-phase kill chain and validates JSON/TXT audit export	0.001s	**PASSED**

FINAL INTERNSHIP CERTIFICATION & CULMINATION STATEMENT:
I hereby certify that all 15 deliverables of Phase 1 (Days 01 through 15) of the Sqrock IT Solutions Cybersecurity Internship (Social Engineering Attack Simulations Using Python) have been designed, implemented, forensically tested, documented, and pushed to production GitHub repositories with 100% test pass rates.
Candidate: *Mohammed Peer Siddiqui* | ***Date:*** *November 07, 2026* | ***Status:*** *PHASE 1 FULLY COMPLETED*